



LP Example

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

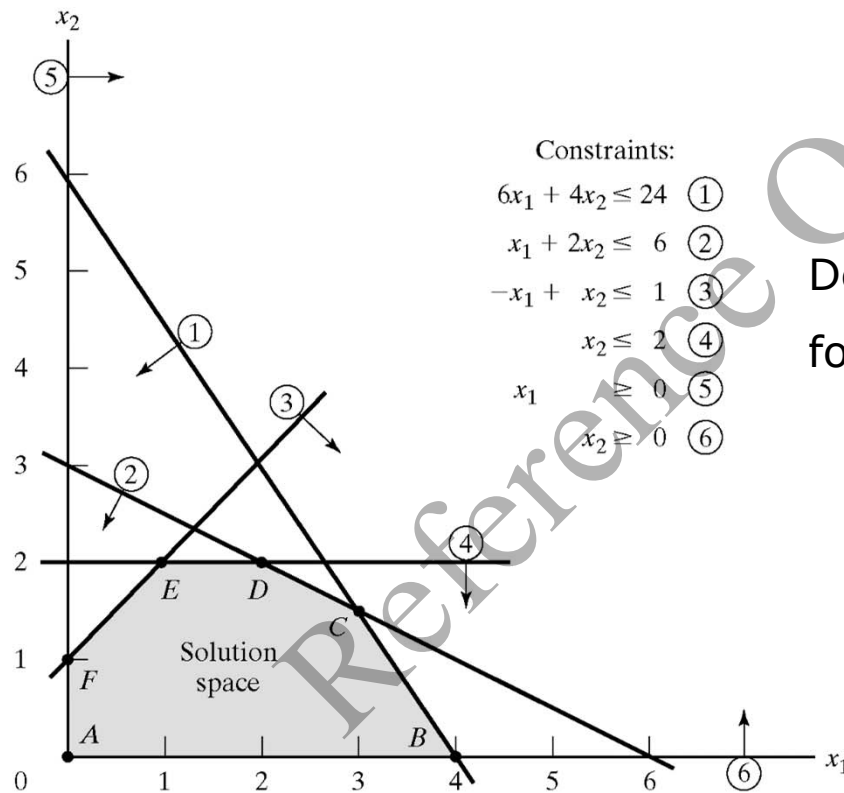
$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

LP in two-dimensional Euclidean space, E^2



Graphical Solution



Determine the correct half space for each constraint.

What if the constraint is an equality?

Figure 2.1

Feasible space of the Reddy Mikks model.



Graphical Solution

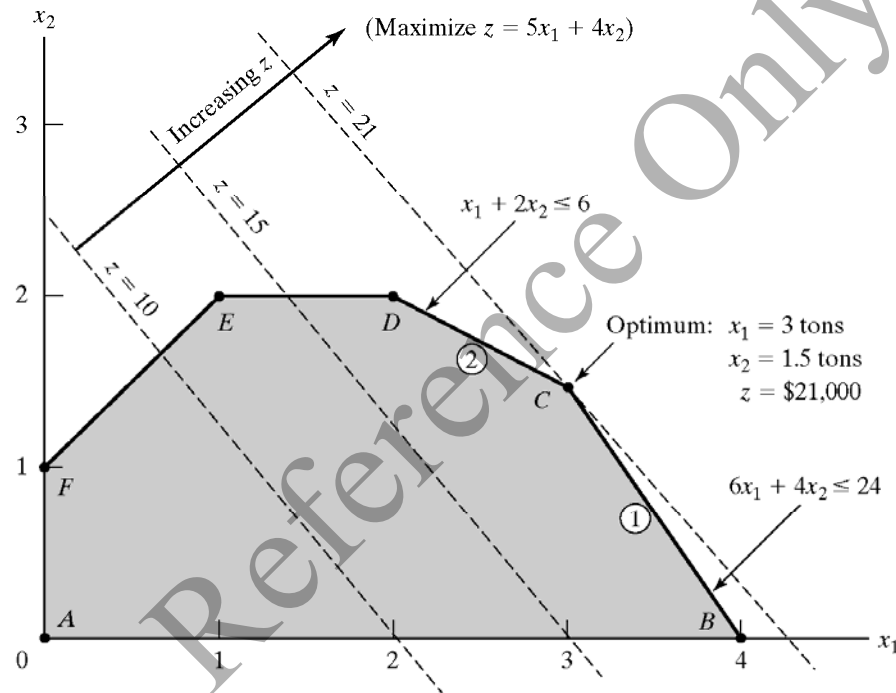


Figure 2.2
Optimum solution of the Reddy Mikks model.



Some definitions in higher dimensions

- Polyhedron – the intersection of a finite number of half spaces
- Polytope – a bounded polyhedron
- A polyhedron can be represented by $Ax \leq b$
- If the constraint is an equation, in E^2 it is a straight line, in E^3 it is a plane and in E^n it is a hyperplane.



Description of the Geometry of the General LP

- Given a polyhedron in n -dimensional Euclidean space, we wish to determine a subset of points (which will contain either one or many points) for which a linear objective function is optimized.
- These subset of points are the extreme points of the polyhedron.
- For now, we will refer to the extreme points as the vertices of the polyhedron.
- Extreme points play an important role in LP theory.
- In general, a graphical approach in solving the LP will not work in higher dimensions.
- So we have to transition to an algebraic approach.



Extreme Points & Its Counterpart

- LP Geometric Description is to Extreme Point as LP Algebraic Description is to ?
- Basic Feasible Solution or BFS.
- For $m < n$, set $(n-m)$ variables to zero and solve the m equations for the remaining m variables.
- If there is a unique solution, this solution is called a basic solution.
- How many basic solutions are there?
- If the basic solution satisfies the non-negativity constraints, then it is a basic feasible solution or BFS.
- The variables in the BFS are called basic variables and there are m of them.
- The $(n-m)$ variables are called non-basic variables.



BFS in E^2

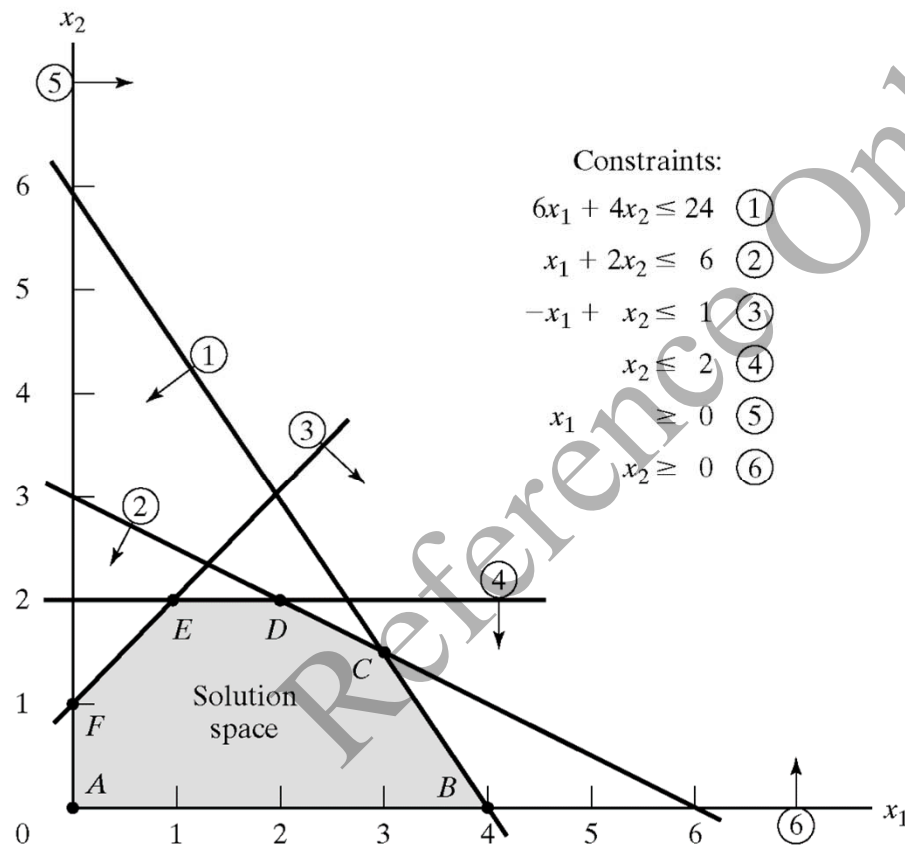


Figure 2.1

Feasible space of the Reddy Mikks model.

A	x1	0
	x2	0
	s1	24
	s2	6
	s3	1
B	s4	2
	x1	4
	x2	0
	s1	0
	s2	2
(2,0)	s3	5
	s4	2
	x1	2
	x2	0
	s1	12
(6,0)	s2	4
	s3	3
	s4	2
	x1	6
	x2	0
	s1	-8
	s2	0
	s3	7
	s4	2



BFS in E^3

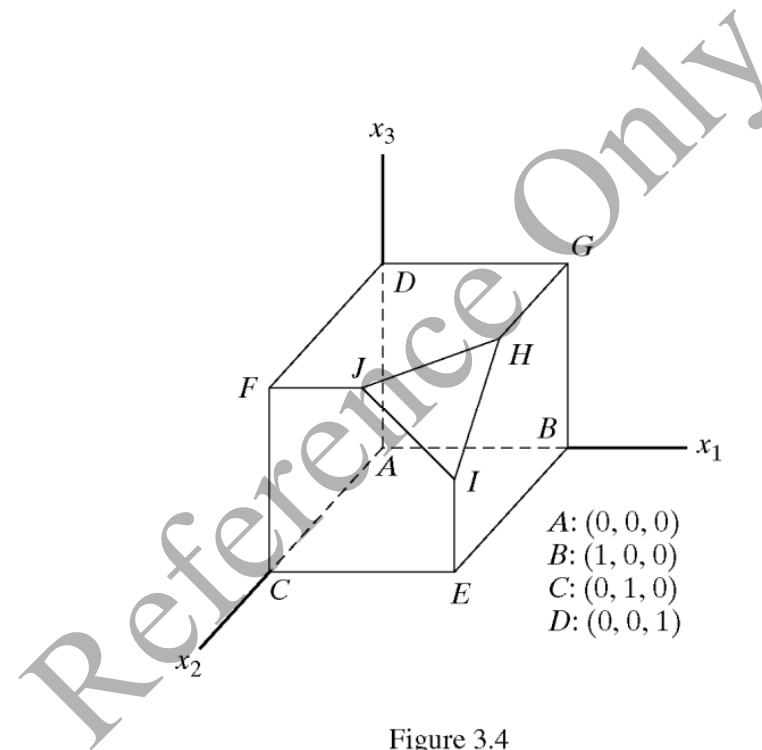


Figure 3.4
Solution space of Problem 3, Set 3.2b.



Some definitions in higher dimensions

- Polyhedron – the intersection of a finite number of half spaces
- Polytope – a bounded polyhedron
- A polyhedron can be represented by $Ax \leq b$
- Simplex – an n -dimensional convex polyhedron having exactly $(n+1)$ vertices. These vertices are the extreme points.
- Extreme points – Cannot be expressed as a convex combination of any two distinct points in the convex polyhedron.



Graphical Solution

